Comprehensive Unload.dll Malware Analysis

Analyst Information

- **Analyst:** Ahmed Emad Nasr
- **Date:** April 5, 2026
- **Target Artifact:** unload.dll
- **Suspected Threat Family:** IcedID (BokBot)

Ahmed Emad Nasr

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1. Executive Summary

This report presents a **static analysis** of the DLL file unload.dll, which exhibits multiple indicators consistent with **IcedID malware behavior**.

Key findings confirm that the binary:

- Is **obfuscated and partially packed**
- Uses **WinHTTP APIs** for outbound communication
- Implements **dynamic API resolution** for evasion
- Contains a **hardcoded C2 domain**
- Likely acts as a **loader for second-stage payloads**

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2. File Metadata & Identification

Attribute	Value
File Name	unload.dll
Original Name	loader_dll_64.dll
File Size	15.94 KiB (16,320 bytes)
SHA-256	A3FA68045D0106D6DB3D43DF6B5997D9034F9F7D2A34148187498E4B504EBF58
Architecture	64-bit (AMD64)
File Type	PE64 DLL
Subsystem	Windows GUI
Compiler	MSVC 19.12
Linker	14.12
Toolset	Visual Studio 2017

Attribute	Value
Timestamp	2022-11-30 20:23:22 UTC
Entropy	4.665
Overlay	Present (Offset: 0x3400, Size: 0x0bc0)

Analysis Notes

- **Moderate entropy (4.665)** indicates possible **partial packing or obfuscation**
- **Overlay presence** strongly suggests **appended encrypted payload/configuration**
- Compilation metadata aligns with **legitimate toolchains**, commonly abused by malware authors

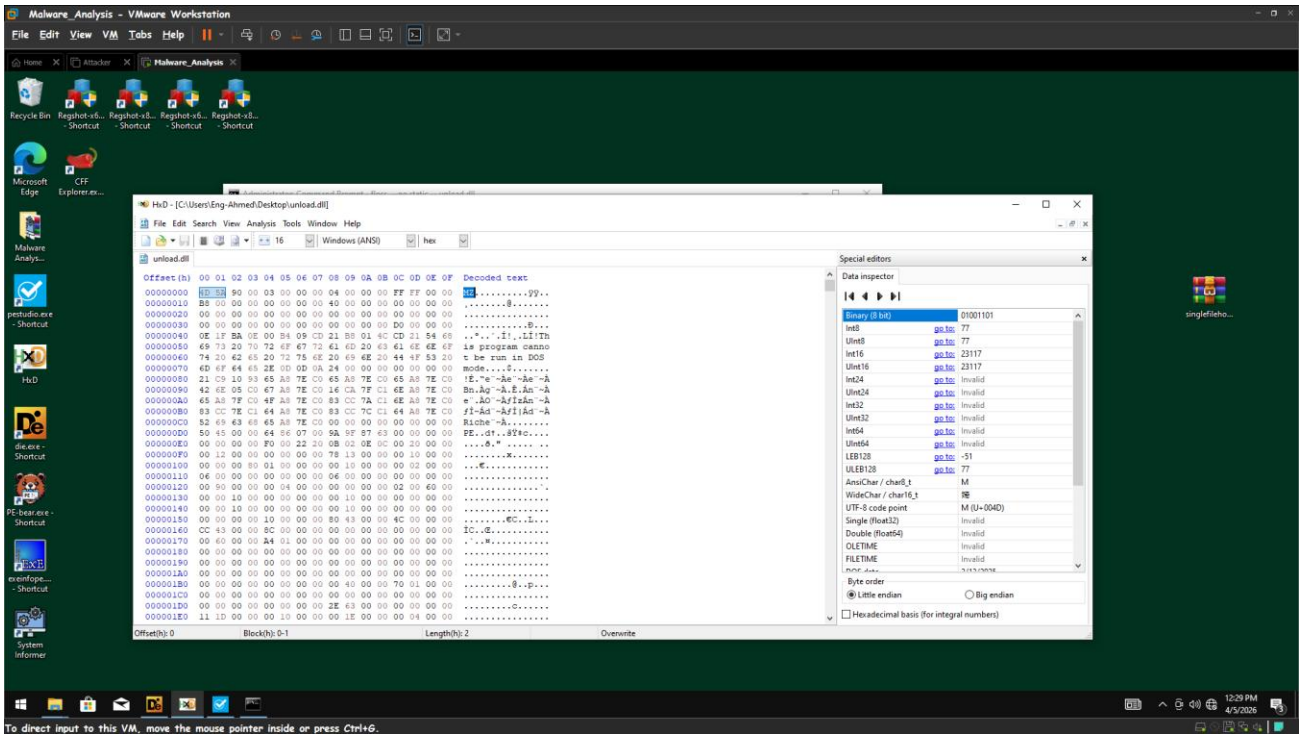
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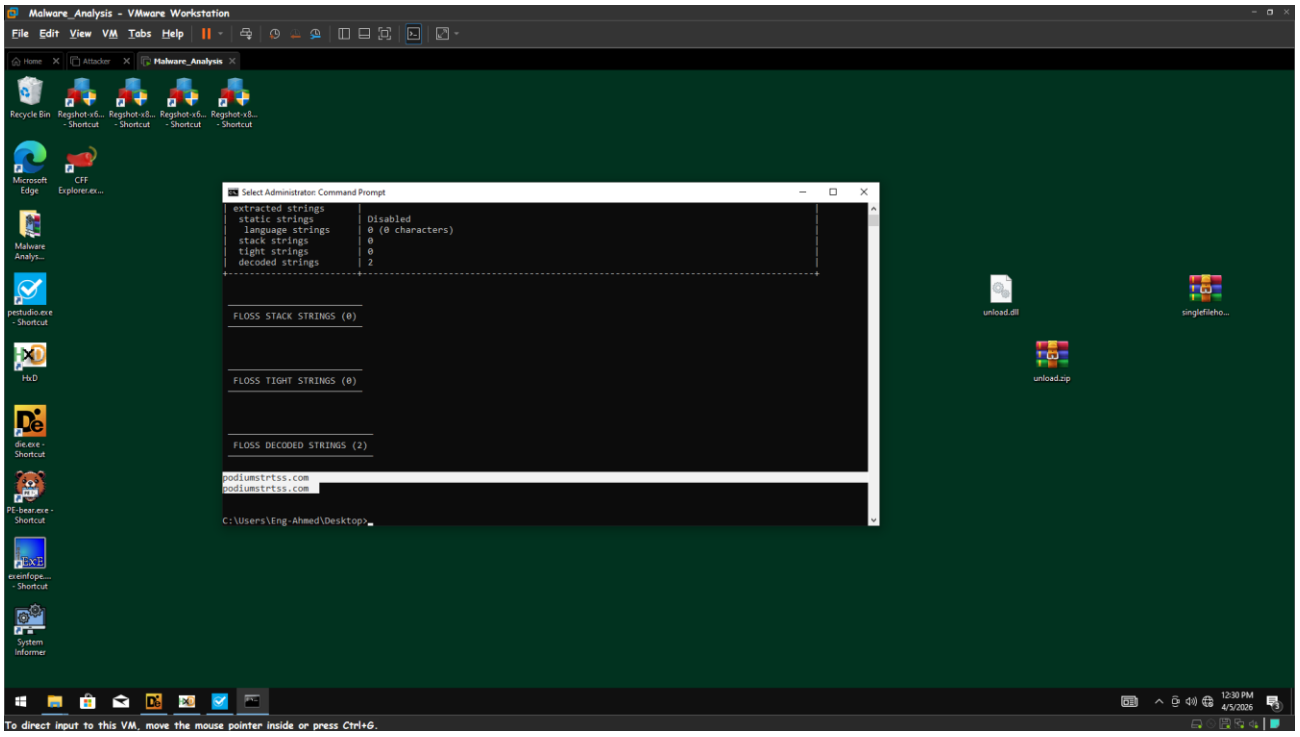
3. PE Section Analysis

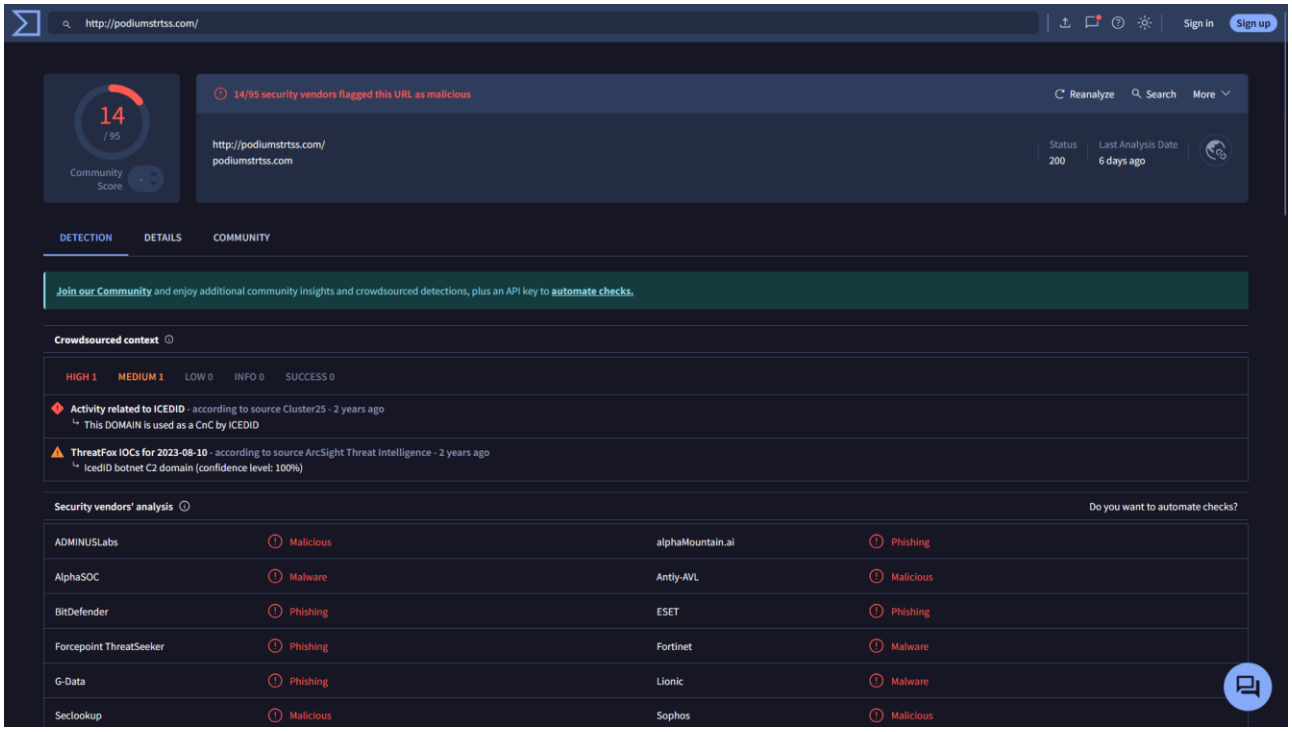
Section	Entropy	Characteristics	Notes
.c	6.153	Executable, Read	Highly suspicious (likely obfuscated code)
.text	0.082	Executable, Read	Extremely small → likely stub
.rdata	4.143	Read-only	Contains imports/strings
.data	N/A	RW	Virtual section
.pdata	3.415	Read	Exception handling
.r	3.399	Read	Non-standard
.d	2.443	RW	Non-standard

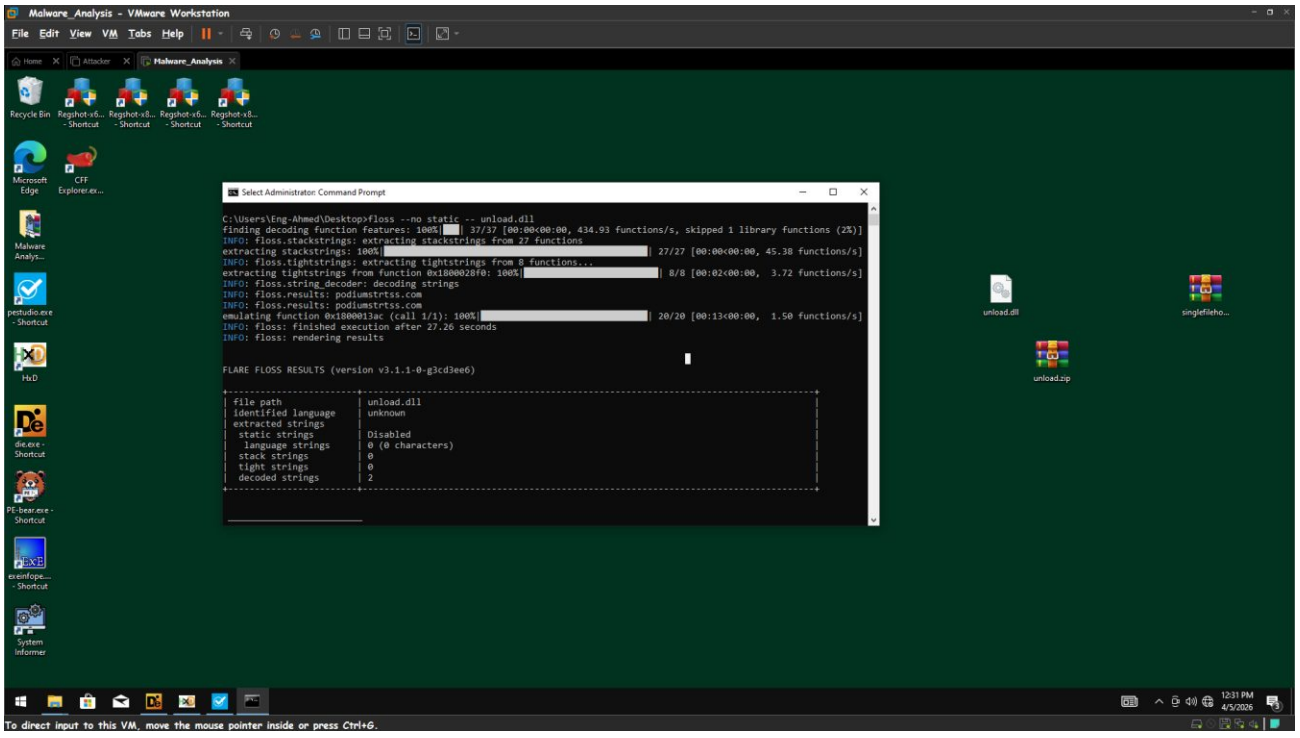
Analysis Notes

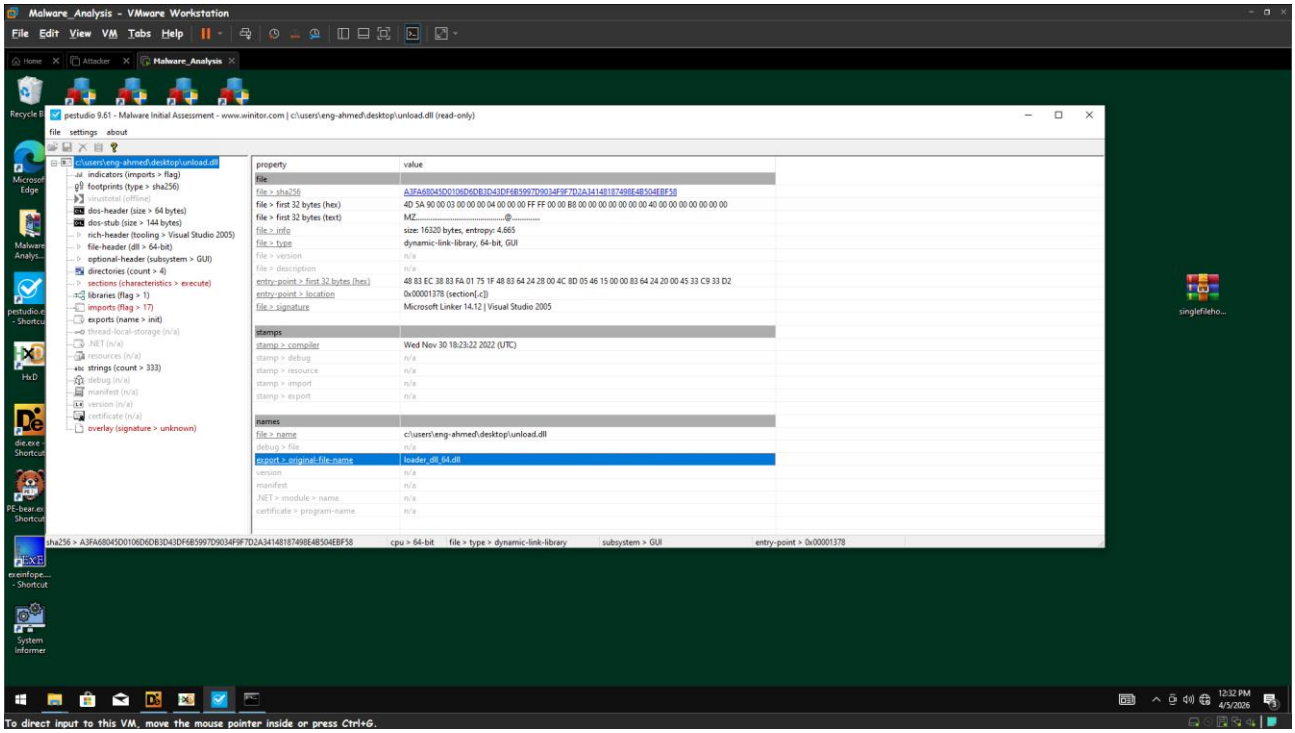
- Use of **non-standard section names** is a strong evasion indicator
- **.c section replacing .text** suggests **manual manipulation or packing**
- **Tiny .text section** indicates **execution redirection to hidden logic**

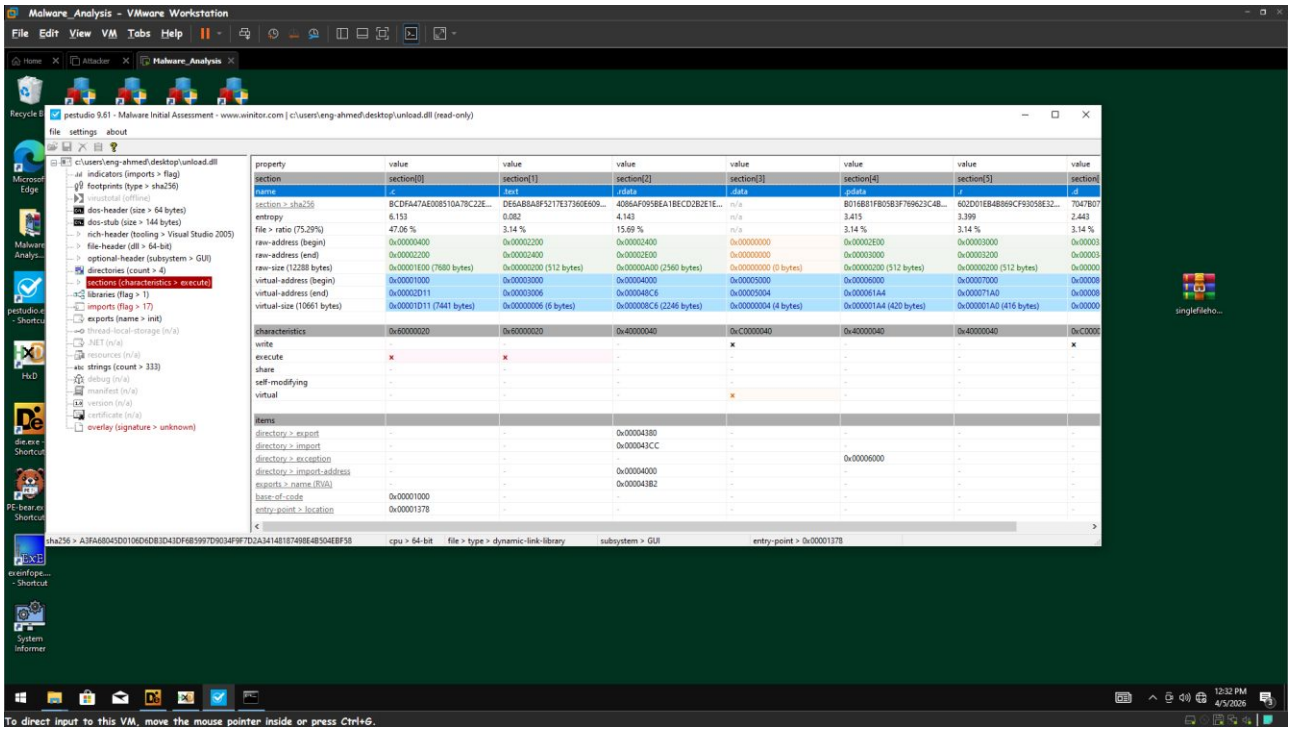

Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
HxD - [C:\Users\Eng-Ahmed\Desktop\unload.dll]
File Edit Search View Analysis Tools Window Help
unload.dll
Special editors
Data inspector
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

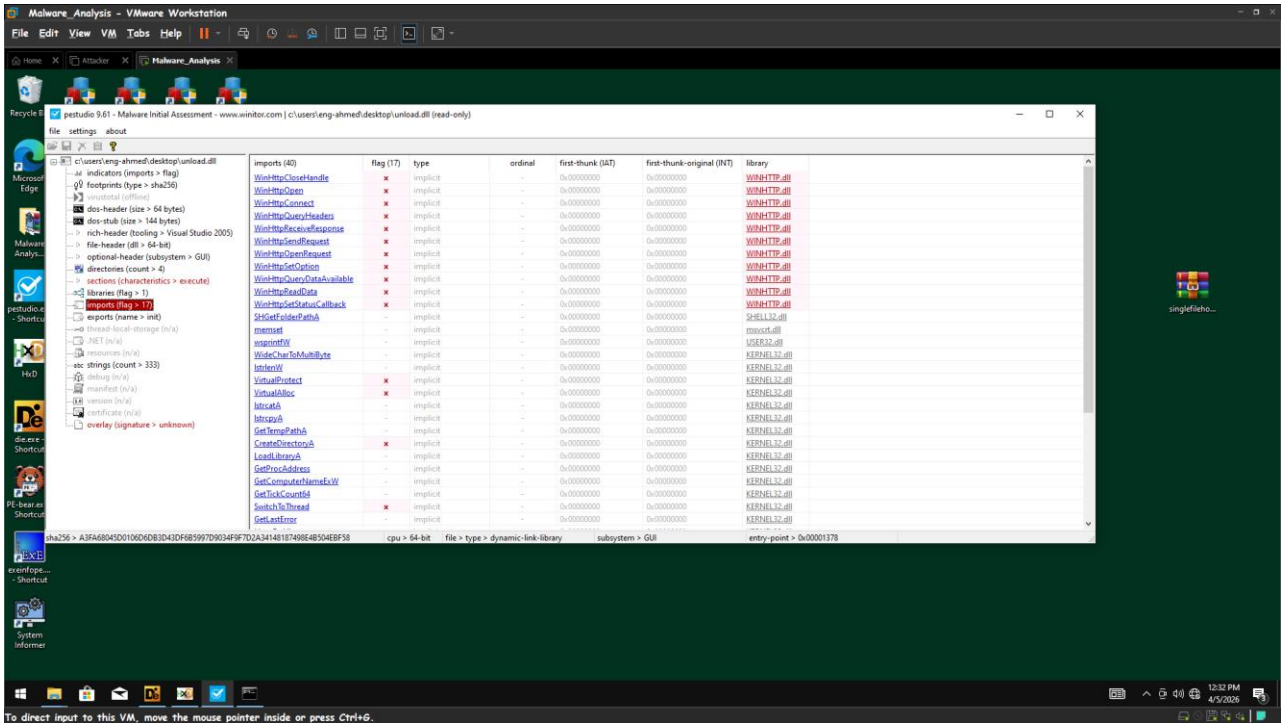

Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
Select Administrator: Command Prompt
FLOSS STACK STRINGS (0)
FLOSS TIGHT STRINGS (0)
FLOSS DECODED STRINGS (2)
C:\Users\Eng-Ahmed\Desktop>
unload.dll
unload.zip
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

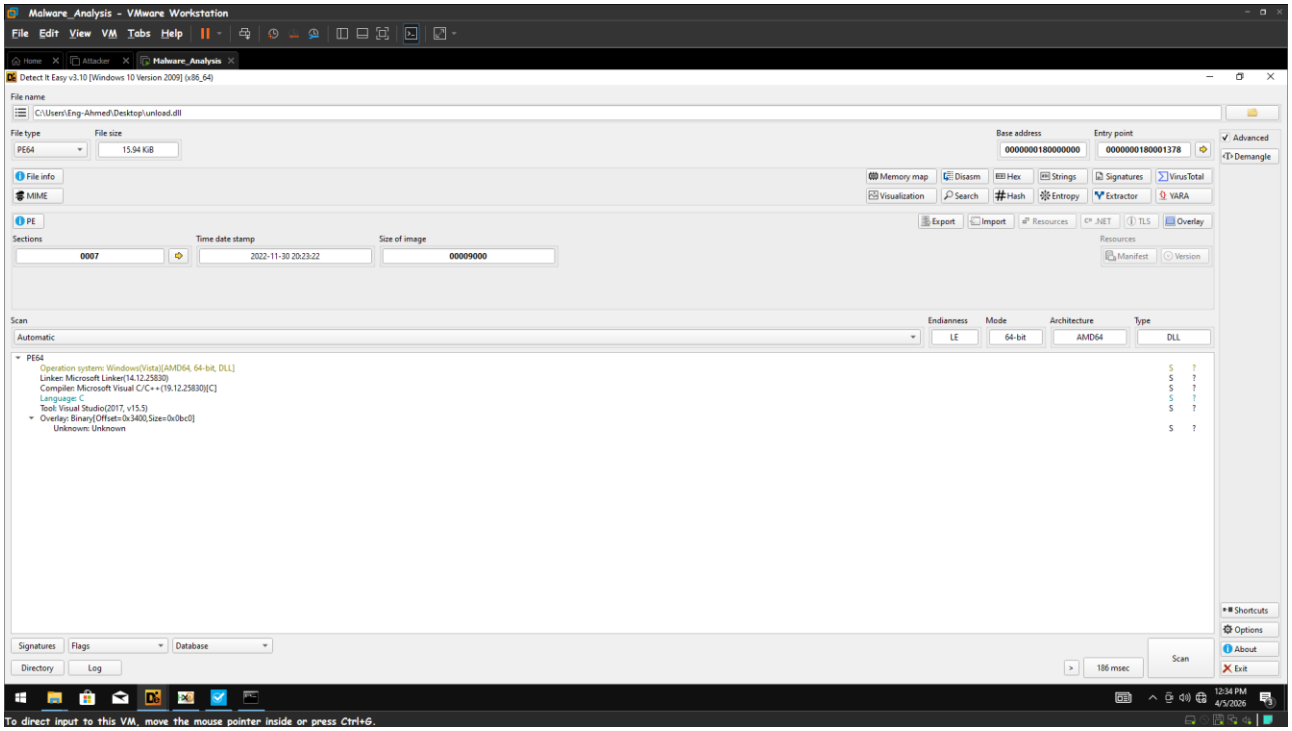

http://podiumstrtss.com/
Sign in
Sign up
14
/ 95
Community Score
14/95 security vendors flagged this URL as malicious
Reanalyze
Search
More
http://podiumstrtss.com/
podiumstrtss.com
Status
200
Last Analysis Date
6 days ago
DETECTION
DETAILS
COMMUNITY
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
Crowdsourced context
HIGH 1
MEDIUM 1
LOW 0
INFO 0
SUCCESS 0
Activity related to ICEDID - according to source Cluster25 - 2 years ago
This DOMAIN is used as a CnC by ICEDID
ThreatFox IOCs for 2023-08-10 - according to source ArcSight Threat Intelligence - 2 years ago
IcedID botnet C2 domain (confidence level: 100%)
Security vendors' analysis
Do you want to automate checks?
ADMINUSLabs
Malicious
alphaMountain.ai
Phishing
AlphaSOC
Malware
Antiy-AVL
Malicious
BitDefender
Phishing
ESET
Phishing
Forcepoint ThreatSeeker
Phishing
Fortinet
Malware
G-Data
Phishing
Lionic
Malware
Seclookup
Malicious
Sophos
Malicious

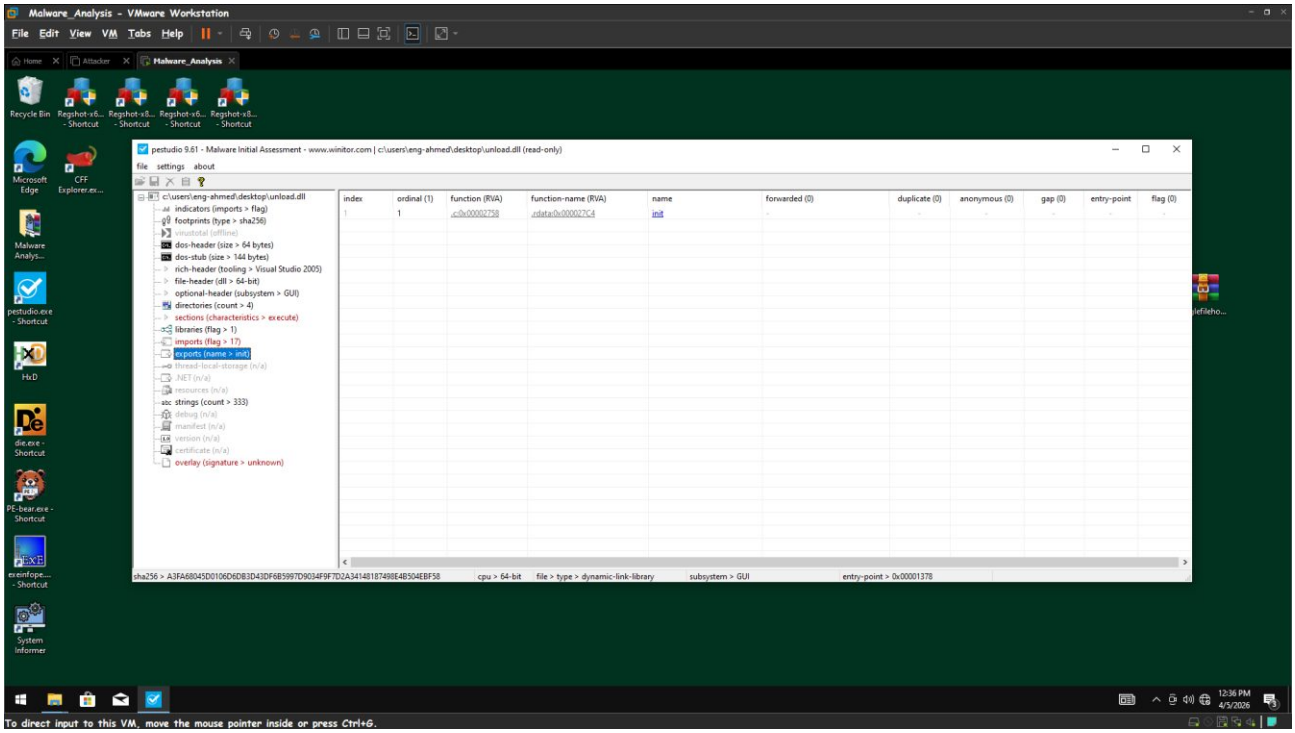

Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
Home
Attacker
Malware_Analysis
Recycle Bin
Microsoft Edge
CFF Explorer.ex...
Malware Analys...
pestudio.exe - Shortcut
HxD
die.exe - Shortcut
PE-bear.exe - Shortcut
exeinfope.... - Shortcut
System Informer
Select Administrator: Command Prompt
C:\Users\Eng-Ahmed\Desktop>floss --no static -- unload.dll
finding decoding function features: 100%| 37/37 [00:00<00:00, 434.93 functions/s, skipped 1 library functions (2%)]
INFO: floss.stackstrings: extracting stackstrings from 27 functions
extracting stackstrings: 100%| 27/27 [00:00<00:00, 45.38 functions/s]
INFO: floss.tightstrings: extracting tightstrings from 8 functions...
extracting tightstrings from function 0x1800028f0: 100%| 8/8 [00:02<00:00, 3.72 functions/s]
INFO: floss.string_decoder: decoding strings
INFO: floss.results: podiumstrtss.com
INFO: floss.results: podiumstrtss.com
emulating function 0x1800013ac (call 1/1): 100%| 20/20 [00:13<00:00, 1.50 functions/s]
INFO: floss: finished execution after 27.26 seconds
INFO: floss: rendering results
FLARE FLOSS RESULTS (version v3.1.1-0-g3cd3ee6)
file path | unload.dll
identified language | unknown
extracted strings
static strings | Disabled
language strings | 0 (0 characters)
stack strings | 0
tight strings | 0
decoded strings | 2
unload.dll
unload.zip
singlefileho...
12:31 PM
4/5/2026
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

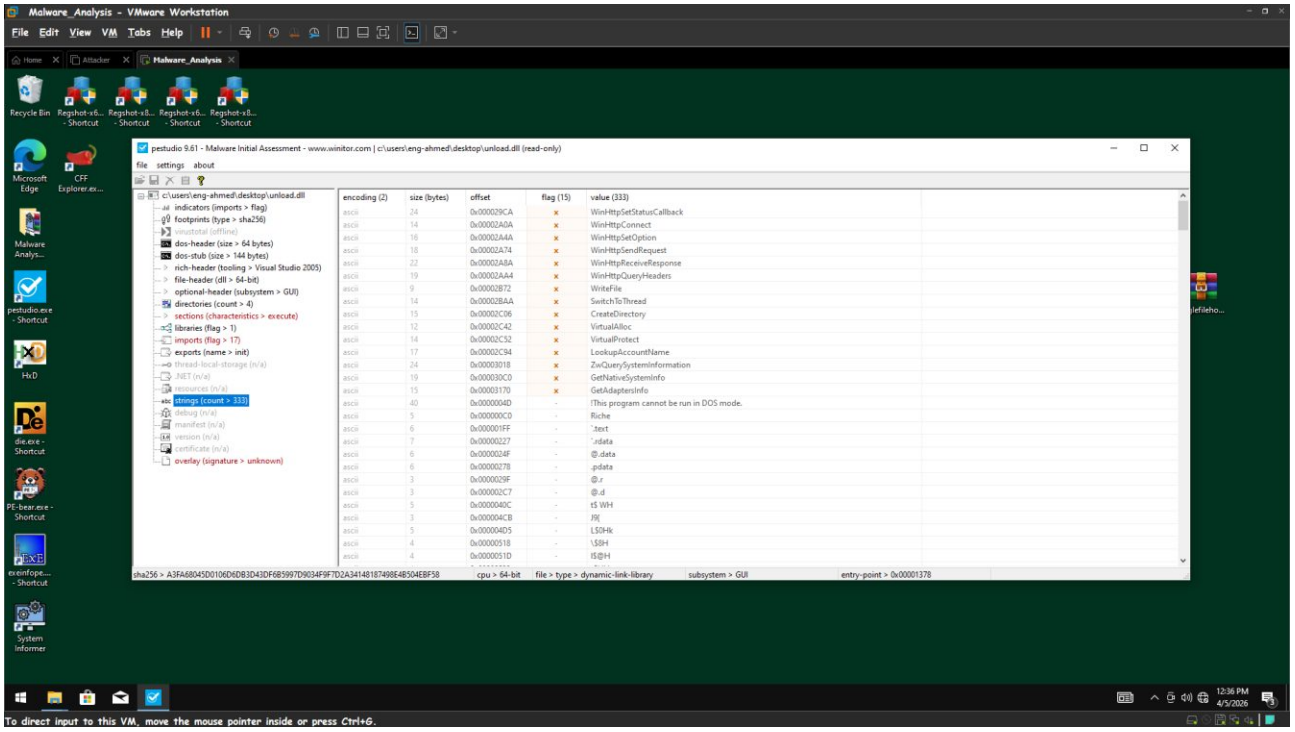

Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\users\eng-ahmed\desktop\unload.dll (read-only)
file settings about
c:\users\eng-ahmed\desktop\unload.dll
indicators (imports > flag)
footprints (type > sha256)
virustotal (offline)
dos-header (size > 64 bytes)
dos-stub (size > 144 bytes)
rich-header (tooling > Visual Studio 2005)
file-header (dll > 64-bit)
optional-header (subsystem > GUI)
directories (count > 4)
sections (characteristics > execute)
libraries (flag > 1)
imports (flag > 17)
exports (name > init)
thread-local-storage (n/a)
.NET (n/a)
resources (n/a)
strings (count > 333)
debug (n/a)
manifest (n/a)
version (n/a)
certificate (n/a)
overlay (signature > unknown)
property
value
file
file > sha256
A3FA68045D0106D6DB3D43DF6B5997D9034F9F7D2A3414818749BE4B504EBF58
file > first 32 bytes (hex)
4D 5A 90 00 03 00 00 00 04 00 00 00 FF FF 00 00 B8 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00
file > first 32 bytes (text)
MZ.....@.....
file > info
size: 16320 bytes, entropy: 4.665
file > type
dynamic-link-library, 64-bit, GUI
file > version
n/a
file > description
n/a
entry-point > first 32 bytes (hex)
48 83 EC 38 83 FA 01 75 1F 48 83 64 24 28 00 4C 8D 05 46 15 00 00 83 64 24 20 00 45 33 C9 33 D2
entry-point > location
0x00001378 (section[.c])
file > signature
Microsoft Linker 14.12 | Visual Studio 2005
stamps
stamp > compiler
Wed Nov 30 18:23:22 2022 (UTC)
stamp > debug
n/a
stamp > resource
n/a
stamp > import
n/a
stamp > export
n/a
names
file > name
c:\users\eng-ahmed\desktop\unload.dll
debug > file
n/a
export > original-file-name
loader_dll_64.dll
version
n/a
manifest
n/a
.NET > module > name
n/a
certificate > program-name
n/a
sha256 > A3FA68045D0106D6DB3D43DF6B5997D9034F9F7D2A3414818749BE4B504EBF58
cpu > 64-bit
file > type > dynamic-link-library
subsystem > GUI
entry-point > 0x00001378
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\users\eng-ahmed\desktop\unload.dll (read-only)
file settings about
c:\users\eng-ahmed\desktop\unload.dll
property
value
value
value
value
value
value
value
section
section[0]
section[1]
section[2]
section[3]
section[4]
section[5]
name
.c
.text
.rdata
.data
.pdata
.rsrc
.d
section > sha256
BCDFA47AE008510A78C22E...
DE6AB8A8F5217E37360E609...
4086AF095BEA1BECD2B2E1E...
n/a
B016B81FB05B3F769623C4B...
602D01EB4B869CF93058E32...
7047B07
entropy
6.153
0.082
4.143
n/a
3.415
3.399
2.443
file > ratio (75.29%)
47.06 %
3.14 %
15.69 %
n/a
3.14 %
3.14 %
3.14 %
raw-address (begin)
0x00000400
0x00002200
0x00002400
0x00000000
0x00002E00
0x00003000
0x00003
raw-address (end)
0x00002200
0x00002400
0x00002E00
0x00000000
0x00003000
0x00003200
0x00003
raw-size (12288 bytes)
0x00001E00 (7680 bytes)
0x00000200 (512 bytes)
0x00000A00 (2560 bytes)
0x00000000 (0 bytes)
0x00000200 (512 bytes)
0x00000200 (512 bytes)
0x00000
virtual-address (begin)
0x00001000
0x00003000
0x00004000
0x00005000
0x00006000
0x00007000
0x00008
virtual-address (end)
0x00002D11
0x00003006
0x000048C6
0x00005004
0x000061A4
0x000071A0
0x00008
virtual-size (10661 bytes)
0x00001D11 (7441 bytes)
0x00000006 (6 bytes)
0x000008C6 (2246 bytes)
0x00000004 (4 bytes)
0x000001A4 (420 bytes)
0x000001A0 (416 bytes)
0x00000
characteristics
0x60000020
0x60000020
0x40000040
0xC0000040
0x40000040
0x40000040
0xC0000
write
execute
share
self-modifying
virtual
items
directory > export
0x00004380
directory > import
0x000043CC
directory > exception
0x00006000
directory > import-address
0x00004000
exports > name (RVA)
0x000043B2
base-of-code
0x00001000
entry-point > location
0x00001378
sha256 > A3FA68045D0106D6DB3D43DF6B5997D9034F9F7D2A3414818749BE4B504EBF58
cpu > 64-bit
file > type > dynamic-link-library
subsystem > GUI
entry-point > 0x00001378
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\users\eng-ahmed\desktop\unload.dll (read-only)
file settings about
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
Detect It Easy v3.10 [Windows 10 Version 2009] (x86_64)
File name
C:\Users\Eng-Ahmed\Desktop\unload.dll
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.





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4. Execution Mechanism (Export Analysis)

Field	Value
Exported Function	init
RVA	0x00002758

Execution Method

rundll32.exe unload.dll,init

Analysis Notes

- Malware relies on **DLL execution via rundll32**
- Exported function acts as **entry point for payload execution**
- Common technique for **living-off-the-land execution (LOLBins)**

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5. Import Address Table (IAT) Analysis

A. Network Communication (WINHTTP.dll)

- WinHttpOpen
- WinHttpConnect
- WinHttpOpenRequest
- WinHttpSendRequest
- WinHttpReceiveResponse
- WinHttpReadData
- WinHttpQueryHeaders
- WinHttpQueryDataAvailable

Conclusion:
Indicates full implementation of **HTTP-based C2 communication**

B. Memory & Execution Control (KERNEL32.dll)

- VirtualAlloc
- VirtualProtect

- LoadLibraryA
- GetProcAddress
- CreateDirectoryA

Conclusion:

- Enables **in-memory execution**
- Supports **dynamic API resolution**
- Suggests **file system interaction capability**

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6. String Deobfuscation (FLOSS Analysis)

Tool Used

floss --no static -- unload.dll

Results

- Decoded function: 0x1800013ac
- Extracted strings: **2**
- Key Indicator:

podiumstrtss.com

Analysis Notes

- Heavy **string obfuscation detected**
- FLOSS confirms **runtime decoding routines**
- Extracted domain acts as **C2 indicator**

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7. Threat Intelligence Correlation

Indicator of Compromise (IOC)

Type Value

Domain podiumstrtss.com

Reputation Analysis

- Detection Rate: **14 / 95 vendors**
- Classified as:
    - **IcedID C2 Infrastructure**
    - **Botnet Communication Endpoint**

Attribution

- Linked to **IcedID (BokBot) malware operations**
- Confirmed by multiple threat intelligence sources

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8. Behavioral Assessment

Based on static analysis, the malware likely performs:

1. **Initial Execution**
    - Triggered via rundll32.exe
2. **C2 Communication**
    - Connects to remote domain via HTTP
3. **Payload Retrieval**
    - Downloads secondary payload
4. **In-Memory Execution**
    - Uses VirtualAlloc + VirtualProtect
5. **Evasion Techniques**
    - Obfuscation
    - Dynamic API resolution
    - Non-standard PE structure

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9. Conclusion

The analyzed DLL is a **malicious loader** strongly associated with the **IcedID malware family**.

It demonstrates:

- **Advanced evasion techniques**
- **Network-based payload delivery**
- **Fileless execution behavior**

This confirms its role as a **Stage-1 loader** designed to:

- Establish persistence
- Retrieve additional malware components
- Execute payloads stealthily in memory

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10. Dynamic Analysis Plan

Monitoring Strategy

1. Sysmon Configuration

- Event ID 1 → Process Creation (rundll32.exe)
- Event ID 3 → Network Connections
- Event ID 11 → File Creation

2. Process Monitoring

- Tool: Process Hacker
- الهدف:
    - Identify **RWX memory regions**
    - Dump decrypted payload

3. Debugging Strategy

- Tool: IDA Pro Debugger
- Breakpoints:
    - WinHttpSendRequest → Capture beacon
    - VirtualProtect → Intercept unpacked payload

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11. Indicators of Compromise (Summary)

Type	Indicator
File Hash	A3FA68045D0106D6DB3D43DF6B5997D9034F9F7D2A34148187498E4B504EBF58
Domain	podiumstrtss.com
Execution	rundll32.exe unload.dll,init

Final Assessment

This sample represents a **high-confidence malicious artifact** associated with **financial malware operations (IcedID)** and should be:

- Blocked at **endpoint and network level**
- Added to **SIEM detection rules**
- Used for **threat hunting across environment**
